

Multi-user Privacy Cooperation Game by Leveraging Users' Service Flexibility

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Abstract—In location-based services (LBSs), it is promising for multiple users to cache and share their Point-of-Interest (PoI) information with each other to reduce overall query frequency and preserve location privacy. Yet most studies on multi-user privacy preservation overlook the opportunity of leveraging service flexibility, where many users are flexible and may add obfuscation to individual LBS query. This paper is the first to study how multiple users cooperate to query with obfuscation against the adversary's optimal inference attack, by leveraging their mutual service flexibility. Unlike the literature, even if a user already finds the shared PoI information useful, we prove it beneficial for him to further query with obfuscated location to confuse the adversary. To save the computational complexity of the max-min adversarial game problem and derive the closed-form solution, we also propose a binary approximate solution, which is proved to guarantee good privacy performance for an average user. Perhaps surprisingly, the user with greater service flexibility should choose to query the LBS with less misreported location, to maximally confuse the adversary. Finally, we numerically compare our optimal and approximate solutions with the existing approaches to show our effective privacy improvement.

I. INTRODUCTION

The prevalence of location-based services (LBSs) brings forth opportunities for service providers to cater to users' needs. LBSs respond to users' queries based on their current locations, and provide useful points of interests (PoIs) nearby. The usage of LBSs, however, might also leak users' private location information, as query data stored in LBS servers may be revealed to advertisers or hacked by malicious attackers [1].

To protect LBS users' privacy, caching-based approaches are considered in [2]–[6]. In particular, [5] proposed multi-user cooperation to cache and share the searched PoI information with each other. Only if the user finds the shared PoI information not useful, he has to query the LBS using his real location, which might be compromised by the adversary. If we allow a user to make multiple queries with extra complexity/overhead, some other works improve the multi-user privacy cooperation via k -anonymity [7]–[9] or l -diversity [10]. Such approaches only work well for a large number k and are also vulnerable to attacks with background knowledge [11].

All these cooperative schemes, however, overlook the opportunity of service flexibility to leverage in users' queries. In practice, many LBS users are flexible in service requirements and only expect the returned PoIs are within walking distances

(e.g., hotels within 1 km) [12], [13]. If a user does not obtain useful PoI information from others and has to query the LBS, he may query with a misreported location to leverage his service flexibility. Furthermore, [2]–[10] simply ask a user to hide his query if useful PoI information is already found in the cache. Yet we prove it beneficial for him to still query with obfuscation to jointly confuse the adversary's attack, gain his privacy, and enlarge the cache coverage.

In this paper, we focus on the fundamental scenario with only one single query for one user in LBS. To maximally preserve the multi-user privacy, we propose to leverage users' service flexibility and take a no-hiding strategy in Section II, which is shown to bring additional privacy performance. After taking the adversary's optimal inference attack into consideration in Section III, we formulate a max-min adversarial game, and simplify it to a linear programming (LP) problem. To save the complexity and derive the closed-form solution, we further propose a binary approximate solution with constant complexity in Section IV. Our approximate solution provides a privacy guarantee and significantly outperforms the state-of-the-art works in Section V. Due to space limit, we put the detailed proofs in the on-line technical report [14].

II. SYSTEM MODEL

In this section, we first introduce our cooperation scheme, and then explain users' model and strategies against the adversary. We consider N LBS users in a set $\mathcal{U} = \{1, 2, \dots, N\}$ sequentially arriving with their real locations $x_1, x_2, \dots, x_N$ in a continuous bounded location set $\mathcal{M}$. Each user $i \in \mathcal{U}$ queries the LBS server for nearby PoI information (e.g., hotels, clubs), by reporting a location $x'_i \in \mathcal{M}$. The LBS server will return PoI information $I(x'_i)$ exactly at x'_i to the user, and user i only finds PoIs within a distance Q_i useful.

With a cooperative cache, each user has access to all previous users' searched PoI information, and stores the reported location x'_i as well as the newly returned PoI information $I(x'_i)$ after his own query (if any). Fig. 1 illustrates our cooperative query process among N users in the following 4 steps.

- Step 1: user i searches the cooperative cache created by previous users, for possible PoI information around x_i .
- Step 2: user i may continue to query the LBS server with a reported location x'_i .
- Step 3: the server returns PoI information $I(x'_i)$ to the user.
- Step 4: user i stores the reported location x'_i and the PoI information $I(x'_i)$ in the cache to benefit latter users.

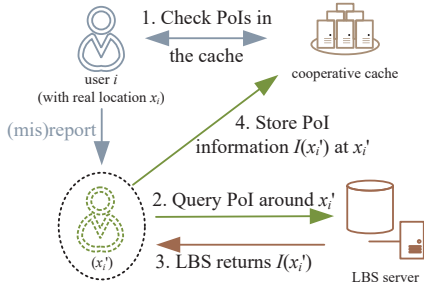


Fig. 1. Users' cooperative LBS query: user i with real location x_i searches the cooperative cache first, finding helpful/unhelpful PoI information. Then he may query the LBS server with a location x'_i and receive the PoI information $I(x'_i)$ at x'_i . To benefit latter users, he stores $(x'_i, I(x'_i))$ in the cache.

Given the random arrival pattern to demand PoIs, each user is at the u -th order ($u = 1, \dots, N$) with probability $\frac{1}{N}$ to observe $u - 1$ former users' query information and receive privacy gain π_u . One can imagine the monotonic relationship with $\pi_u \leq \pi_{u+1}$. After taking the expectation, each user's expected privacy gain is

$$\Pi = \frac{1}{N} \sum_{u=1, \dots, N} \pi_u. \quad (1)$$

Particularly, without loss of generality, we suppose user i is the i -th among N users to arrive and he may or may not benefit from the previously shared information, leading to different reporting strategies in Step 2. Similar to [15], each user faces a strict service quality constraint to be only interested in PoIs around him. Thus, user i benefits from the cache if he finds any user j 's shared $I(x'_j)$ within distance Q_i , i.e., $d(x_i, x'_j) \leq Q_i$, where $d(\cdot)$ measures the Euclidean distance. Mathematically, let

$$\mathcal{X}_i^{IN}(x'_{i-1}) = \{x \in \mathcal{M} \mid \min_{j=1, \dots, i-1} d(x, x'_j) \leq Q_i\} \quad (2)$$

denote the location set for user i 's real location x_i within which nearby PoI information (within distance Q_i) is already stored by previous users' queries $\mathbf{x}'_{i-1} = (x'_1, \dots, x'_{i-1})$. Let

$$\mathcal{X}_i^{OUT}(x'_{i-1}) = \{x \in \mathcal{M} \mid \min_{j=1, \dots, i-1} d(x, x'_j) > Q_i\}$$

denote the location set for user i 's real location x_i within with nearby PoI information is uncovered by the cache. If $x_i \in \mathcal{X}_i^{IN}$, user i 's service constraint is satisfied and he will find the cached PoIs useful. Otherwise, if $x_i \in \mathcal{X}_i^{OUT}$, user i 's reporting strategy should be designed to meet user i 's service quality constraint: $d(x_i, x'_i) \leq Q_i, \forall x_i \in \mathcal{X}_i^{OUT}$. Note that the first-arriving user 1 finds the cache empty with $\mathcal{X}_1^{IN} = \emptyset$.

To determine the reported location x'_i , we use a conditional probabilistic distribution $f_i(x'_i|x_i)$ to denote user i 's reporting strategy, i.e., a mapping probability from $x_i \in \mathcal{M}$ to $x'_i \in \mathcal{M}$. Depending on whether the cached PoIs are useful, we can then generally define user i 's reporting strategy to LBS as

$$f_i(x'_i|x_i, \mathbf{x}'_{i-1}) = \begin{cases} f_i^{in}(x'_i|x_i, \mathbf{x}'_{i-1}), & \text{if } x_i \in \mathcal{X}_i^{IN}, \\ f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1}), & \text{if } x_i \in \mathcal{X}_i^{OUT}. \end{cases} \quad (3)$$

Given that the adversary will infer user i 's real location as $\hat{x}_i(x'_i|\mathbf{x}'_{i-1})$ after observing user i 's reporting x'_i as well as all former users' queries $\mathbf{x}'_{i-1}$, we define user i 's privacy gain as

the adversary's expected error in the inference attack:

$$\begin{aligned} \pi_i &= \int_{x'_i} \int_{x_i \in \mathcal{X}_i^{IN}} \psi_i(x_i) f_i^{in}(x'_i|x_i, \mathbf{x}'_{i-1}) d(x_i, \hat{x}_i) dx_i dx'_i \\ &+ \int_{x'_i} \int_{x_i \in \mathcal{X}_i^{OUT}} \psi_i(x_i) f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1}) d(x_i, \hat{x}_i) dx_i dx'_i \\ &= \int_{x_i, x'_i} \Pr(x_i, x'_i|\mathbf{x}'_{i-1}) d(x_i, \hat{x}_i) dx_i dx'_i, \end{aligned} \quad (4)$$

where $\psi_i(x_i)$ is user i 's location distribution. Later we will analyze the adversary's optimal inference and show $\hat{x}_i$ is a function of $f_i(x'_i|x_i, \mathbf{x}'_{i-1})$ in Section III. Based on it, we formulate the privacy cooperation problem for each user.

III. OPTIMAL PRIVACY COOPERATION AGAINST ADVERSARIAL INFERENCE ATTACK

In this section, we model the strategic interaction between users and the adversary as a two-stage adversarial game, where users make the query move first against the adversary's inference attack. We formulate each user's privacy-preservation as a max-min optimization problem, and simplify it as an LP.

A. Adversary's Optimal Inference Attack

To provide a robust privacy defence for the user, we consider the worst-case of the adversary's strongest information. By hacking into the cache, any cooperative peers and/or the historical data from the LBS platform, we assume that the adversary has full knowledge of the cached data $\mathbf{x}'_{i-1}$, each user's privacy objective function as in (4), and their location distributions $\psi_i(x_i)$ ($i = 1, \dots, N$) (i.e., a Bayesian setting [16], [17]). Thus it can estimate users' reporting strategies $f_i(x'_i|x_i, \mathbf{x}'_{i-1})$ on their behalves, and launch the optimal Bayesian inference attack of each user's real location.

In particular, after observing user i 's reported location x'_i as well as the cached PoI information, the posterior distribution of user i 's real location x_i to the adversary is a function of $i - 1$ former users' queries $\mathbf{x}'_{i-1}$:

$$\begin{aligned} \Pr(x_i|x'_i, \mathbf{x}'_{i-1}) &= \frac{\Pr(x_i, x'_i|\mathbf{x}'_{i-1})}{\Pr(x'_i|\mathbf{x}'_{i-1})} \\ &= \frac{\psi_i(x_i) f_i(x'_i|x_i, \mathbf{x}'_{i-1})}{\int_{x_i \in \mathcal{M}} \psi_i(x_i) f_i(x'_i|x_i, \mathbf{x}'_{i-1}) dx_i}, \end{aligned} \quad (5)$$

where the denominator is the integral over all the location region $\mathcal{M}$, including the strategy $f_i^{in}(x'_i|x_i, \mathbf{x}'_{i-1})$ for $x_i \in \mathcal{X}_i^{IN}(\mathbf{x}'_{i-1})$ and $f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1})$ for $x_i \in \mathcal{X}_i^{OUT}(\mathbf{x}'_{i-1})$. To minimize the inference error to user i 's real location x_i , the adversary's optimal inference problem for user i is given as

$$\min_{\hat{x}_i} \int_{x_i} \Pr(x_i|x'_i, \mathbf{x}'_{i-1}) d(\hat{x}_i, x_i) dx_i. \quad (6)$$

As a special case, for the first user out of N users to arrive with $\mathcal{X}_1^{IN} = \emptyset$ and $\mathcal{X}_1^{OUT} = \mathcal{M}$, (5) reduces to

$$\Pr(x_1|x'_1) = \frac{\Pr(x_1, x'_1)}{\Pr(x'_1)} = \frac{\psi_1(x_1) f_1^{out}(x'_1|x_1)}{\int_{x_1 \in \mathcal{M}} \psi_1(x_1) f_1^{out}(x'_1|x_1) dx_1}.$$

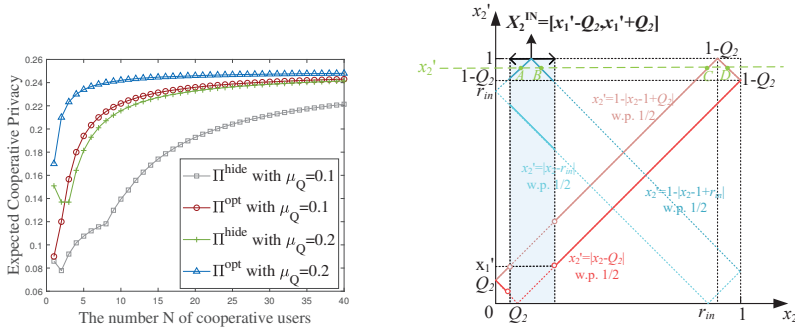


Fig. 2. Performance comparison between our Fig. 3. Example of user 2's optimal query strategy to Problem (8) and the typical reference $f_2(x'_2|x_2)$ with r_{in}^* from (11) in Proposition 2 and $r_{out}^* = Q_2$, given $Q_2 < x'_1 \leq \frac{1}{2}$. Q_i follows the truncated normal distribution with mean $\mu_Q = 0.1, 0.2$, $\sigma = 0.1$.

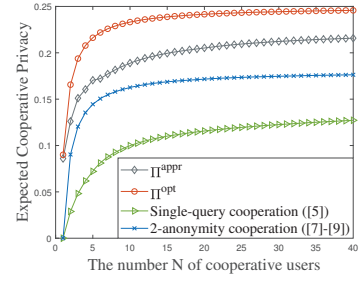


Fig. 4. Performance evaluation of the approximate solution with the optimal strategy and schemes in the literature. Q_i follows the truncated normal distribution with $\mu = 0.1$, $\sigma = 0.1$ and $r_{in} = 0.5$.

B. Max-min Problem Formulation and Simplification

Based on the adversary's strategies above, we formally model the strategic interaction between user i and the adversary as a two-stage game:

- In Stage I, user i makes a probabilistic strategy $\mathbf{f}_i(x'_i|x_i) = \{f_i^{in}(x'_i|x_i), f_i^{out}(x'_i|x_i)\}$ on how to report location to the LBS server to maximize his privacy gain in (4), considering whether he finds the cached PoI information useful or not.
- In Stage II, the adversary infers the user i 's location as $\hat{x}_i$ after observing the reported location x'_i as well as former users' reports $\mathbf{x}'_{i-1}$, to minimize its inference error in (6).

Note that there is no conflict or strategic interaction between any two users, as they sequentially demand PoI information at different time stages. In other words, all users follow the design to play against the adversary as a coalition. To combine the two stages, we formally formulate each user i 's privacy protection against the adversary's optimal inference as

$$\begin{aligned} \max_{x'_i} \int_{x'_i} \Pr(x'_i|\mathbf{x}'_{i-1}) \min_{\hat{x}_i} \int_{x_i} \Pr(x_i|x'_i, \mathbf{x}'_{i-1}) d(\hat{x}_i, x_i) dx_i dx'_i \\ = \int_{x'_i} \min_{\hat{x}_i} \int_{x_i \in \mathcal{M}} \psi_i(x_i) f_i(x'_i|x_i, \mathbf{x}'_{i-1}) d(\hat{x}_i, x_i) dx_i dx'_i \\ \text{s.t. } f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1}) = 0, \\ \forall (x_i, x'_i) \in \{(x_i, x'_i) | x_i \in \mathcal{X}_i^{OUT}, x'_i \in \mathcal{M} : d(x'_i, x_i) > Q_i\}, \\ \int_{x'_i} f_i(x'_i|x_i, \mathbf{x}'_{i-1}) dx'_i = 1, \forall x_i \in \mathcal{M}, \\ \text{var} : f_i(x'_i|x_i, \mathbf{x}'_{i-1}) \text{ as in (3)}. \end{aligned} \quad (7)$$

The objective is given from user i 's privacy π_i in (4) and the adversary's inference problem in (6). The first constraint ensures that user i out of cache coverage ($x_i \in \mathcal{X}_i^{OUT}$) will not report a location x'_i with more than distance Q_i away from his real location x_i to satisfy his service constraint.

The region $\mathcal{M}$ is continuous and it is difficult to solve two continuous variable functions $f_i^{in}(x'_i|x_i, \mathbf{x}'_{i-1})$ and $f_i^{out}(x'_i|x_i, \mathbf{x}'_{i-1})$ in (7). We propose the following alternative.

Lemma 1. By equally partitioning the location set $\mathcal{M}$ into M grids, we reformulate the the objective in Problem (7) as

$$\max_{x'_i} \sum_{\hat{x}_i} \min_{x_i \in \mathcal{M}} \psi_i(x_i) f_i(x'_i|x_i, \mathbf{x}'_{i-1}) d(\hat{x}_i, x_i), \quad (8)$$

then the problem can be solved by an LP with total complexity $O(NM^7)$ in the number N of cooperative users.

Recall that in the multi-user privacy preservation schemes in [5]–[10], users query the LBS server only if the target PoI information is not found in the cache, while in our problem user i still queries with strategy f_i^{in} as in (3), even if he has already got the needed PoI information. Following their hiding-from-the-LBS idea, we have $f_i^{in} = \emptyset$ as there is no query for $x_i \in \mathcal{X}^{IN}$, and the adversary will infer randomly in the region $\mathcal{X}^{IN}$ without any observation. We can similarly formulate a max-min problem for the hiding scheme as in (7), yet the decision variables only include f_i^{out} for $x_i \in \mathcal{X}_i^{OUT}$. Let Π^{hide} denote the expected privacy gain per user under the hiding-from-the-LBS scheme, and Π^{opt} denote our solution's counterpart by always querying the LBS server strategically. Next we compare them to tell our solution's advantage.

Lemma 2. $\Pi^{opt} > \Pi^{hide}$ always holds and it is always beneficial for each user to strategically query the LBS.

Even if the user already gets help from the shared PoI information, querying the LBS server strategically helps further confuse the adversary. Fig. 2 quantitatively compares our Π^{opt} (by solving Problem (8)) with Π^{hide} to tell obvious privacy performance gain for any user number N , where Q_i 's for all users follow the same distribution (e.g., the truncated normal distribution with the minimum value $Q_i \geq 0$). Overall, all curves are increasing in N and the privacy gain increases in the mean value of Q_i . Yet Π^{hide} decreases from $N = 1$ to 3, since the second/third user's hiding exposes the user's narrowed location region in $\mathcal{X}_i^{IN}$ to be around the former user's and helps the adversary infer more accurately.

IV. BINARY APPROXIMATE SOLUTION

No matter in Problem (7) or (8), we can only numerically derive the solution for multi-user privacy cooperation. Even for Problem (8), it still takes time to return the solution for a large number of cooperative users. To make the problem scalable and provide clean engineering insights, this section presents our binary approximation solution in closed-form. Without much loss of generality, we narrow the users' location set $\mathcal{M}$ to a normalized one-dimensional(1D) linear line $[0, 1]$, though our analysis and solution can also be extended to two-dimension(2D). We also assume that each user can be equally located at any point in $\mathcal{M}$ by following the uniform distribution with PDF $\psi_i(x_i = x) = 1$ ($x \in [0, 1]$).

Instead of using continuous f_i^{in} and f_i^{out} over the location set $\mathcal{M}$ as in (3), we narrow each of them to the randomization of only two location points around x_i .

Definition 1 (Binary approximation for LBS query). *In the 1D line domain $[0, 1]$, if a user i finds the cached PoI information useful (i.e., $x_i \in \mathcal{X}_i^{IN}(x'_{i-1})$), his reporting strategy f_i^{in} is to randomize two points at fixed distances on the two sides of his real location x_i :*

$$f_i^{in}(x'_i|x_i, x'_{i-1}) = \begin{cases} \frac{1}{2}, & \text{if } x'_i = |x_i - r_{in}|, \\ \frac{1}{2}, & \text{if } x'_i = 1 - |x_i - 1 + r_{in}|, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Otherwise if $x_i \in \mathcal{X}_i^{OUT}(x'_{i-1})$,

$$f_i^{out}(x'_i|x_i, x'_{i-1}) = \begin{cases} \frac{1}{2}, & \text{if } x'_i = |x_i - r_{out}|, \\ \frac{1}{2}, & \text{if } x'_i = 1 - |x_i - 1 + r_{out}|, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Such binary approximation is the simplest but fundamental way to replace the complicated randomization f_i^{in} and f_i^{out} . We decide different r_{in} and r_{out} to adaptively fit the two cases of using cache. Next we will analyze how to optimally design r_{in} and r_{out} in (9) and (10) from the first and latter users' points of view, and then prove that our approximation solution guarantees close-to-optimal performance for an average user.

A. The First User's Random Query Strategy

First, we discuss the optimal randomized strategy for the first-arriving user, who observes an empty cache (i.e., $\mathcal{X}^{IN} = \emptyset$) and acts as an independent user to decide the reporting strategy. Forecasting the optimal inference attack analyzed as in Section III-A, we proactively design r_{out} to maximize his expected privacy gain.

Proposition 1. *The first-arriving user will randomly misreport his location with $r_{out}^* = \min(Q_1, \frac{1}{2})$ in (10). His maximum expected privacy gain is $\pi_1 = \min(Q_1 - Q_1^2, \frac{1}{4})$.*

B. Latter Users' Random Query Strategy

For any user arriving later, he may take different strategies r_{in} in (9) and r_{out} in (10) adaptively depending on whether he benefits from the shared information. For ease of exposition, we focus on user 2 who is the second to arrive to explain, and

similar analysis can be applied to latter users. We will numerically solve latter users' reporting strategy and corresponding privacy gain in the expected sense later in Section V-A.

Given the first user 1's query x'_1 from Proposition 1, if $x_2 \in \mathcal{X}_2^{OUT}(x'_1) = \{x_2 \in \mathcal{M} | d(x'_1, x_2) > Q_2\}$, user 2 does not benefit from user 1's PoI sharing, then we assume user 2 follows strategy $f_2^{out}(x'_2|x_2)$ in (10) with $r_{out} = r_{out}^* = \min(Q_1, \frac{1}{2})$ as in Proposition 1. If $x_2 \in \mathcal{X}_2^{IN}(x'_1) = \{x_2 \in \mathcal{M} | d(x'_1, x_2) \leq Q_2\}$, user 1's query x'_1 is helpful to user. Following Lemma 2, user 2 will still misreport $x'_2 = |x_2 - r_{in}|$ and $x'_2 = 1 - |x_2 - 1 + r_{in}|$ randomly with a distance r_{in} away in (9). Next we optimize r_{in} for maximizing user 2's expected privacy gain π_2 . To avoid the trivial case that user 2 has sufficient flexibility Q_2 in the first constraint of (7) or (8) to arbitrarily misreport his location without service loss, we consider a challenging case with $Q_2 < 1/11$ here. Notice that we assume a small value range for Q_2 under the least flexible case. The analysis can be extended for a larger Q_2 with a better privacy performance (see the privacy increases in Q_i 's in Fig. 2 for an illustration).

Proposition 2. *Even if user 2 finds the PoI information shared by user 1 useful (i.e., $d(x'_1, x_2) \leq Q_2$), it is optimal for him to misreport x'_2 with distance r_{in}^* to the LBS server:*

$$r_{in}^*(x'_1) = \begin{cases} 1 - Q_2, & \text{if } 0 \leq x'_1 \leq Q_2, \\ 1 - x'_1, & \text{if } Q_2 < x'_1 \leq \frac{1}{2}, \\ x'_1, & \text{if } \frac{1}{2} < x'_1 \leq 1 - Q_2, \\ 1 - Q_2, & \text{if } 1 - Q_2 < x'_1 \leq 1, \end{cases} \quad (11)$$

which is a function of user 1's earlier query x'_1 .

This binary approximation solution with misreporting distances r_{in}^* in (11) and $r_{out}^* = Q_2$ are given in closed-form, thus we manage to reduce the computational complexity of the user strategy from $O(NM^7)$ in Problem (8) to $O(1)$.

Notice that r_{in}^* in (11) is non-increasing in the service flexibility Q_2 . This tells that the user with greater service flexibility should choose to query the LBS with less misreported location. To explain this counter-intuitive result, we first note that the service constraint is actually relaxed here when designing r_{in}^* . Then user 2's query x'_2 for $x_2 \in \mathcal{X}_2^{IN}(x'_1)$ is just to bring maximal confusion to the adversary in the reserve way of thinking, who naturally believes that user 2 will leverage his greater Q_2 to misreport x'_2 further away from the real x_2 .

To better explain the implication of r_{in}^* in (11), we present Fig. 3 giving an example of user 2's reporting strategy and other three cases are similar. In the example case of user 1's realized query $Q_2 < x'_1 \leq \frac{1}{2}$, user 2 has a shaded region $\mathcal{X}_2^{IN}(x'_1)$ within which his target PoI can be covered by user 1's query. If $x_2 \in \mathcal{X}_2^{IN}(x'_1)$, user 2's service constraint is relaxed and he will report on the two sides of x_2 randomly with $f_2^{in}(x'_2|x_2, x'_1)$ in (9) and r_{in}^* in (11). Each blue line ($x'_2 = |x_2 - r_{in}|$ or $x'_2 = 1 - |x_2 - 1 + r_{in}|$) is randomly adopted with an equal probability $\frac{1}{2}$. If $x_2 \in \mathcal{X}_2^{OUT}(x'_1)$, user 2 cannot benefit from the shared information and he will report with $f_2^{out}(x'_2|x_2, x'_1)$ in (10) and $r_{out} = Q_2$ to satisfy the service requirement. Each red line ($x'_2 = |x_2 - Q_2|$ or $x'_2 =$

$1 - |x_2 - 1 + Q_2|$) is randomly adopted with probability $\frac{1}{2}$.

We can observe that user 2 chooses a large r_{in}^* (close to 1) to optimize his expected privacy over $x_2 \in [0, 1]$. In this example, considering that the adversary observes user 2's query x_2' as denoted by the green dash line in Fig. 3, the observation x_2' has four intersections A, B, C, D with user 2's reporting strategy. This leads to four possible real locations with equal probabilities $\frac{1}{2}$: the adversary is not sure whether user 2 is in region $\mathcal{X}_2^{IN}$ (points A, B) or in region $\mathcal{X}_2^{OUT}$ (points C, D). Then the adversary will have to compromise among these four possible locations to optimally attack as in (6). This is why the user still queries with misreported locations instead of hiding even if he gets useful information in the cache (as stated in Lemma 2): the f^{in} strategy works together with f^{out} and brings more confusion to the adversary.

Given the latter user 2's reporting strategy is based on the former user 1's reports x_1' , the following Corollary 1 gives the impact of x_1' to user 2's privacy gain.

Corollary 1. *User 2 (with cooperative order 2)'s expected privacy $\pi_2^{appr}(x_1')$ in the approximate cooperative strategy reaches the minimum when $x_1' = \frac{1}{2}$.*

Corollary 1 shows that an earlier query x_1' located at the centre of the region leads to the worst privacy gain π_2^{appr} for user 2. The reason behind is that when x_1' is near the centre of the location set, user 2 can make less confusion to the adversary by misreporting in $\mathcal{X}_2^{IN}(x_1')$. For example, as shown in Fig. 3, when x_1' is larger, user 2's reporting strategy within $\mathcal{X}_2^{IN}(x_1')$ in blue lines will be closer to those within $\mathcal{X}_2^{OUT}(x_1')$ in red lines. This leads to a smaller privacy gain with less confusion to the adversary.

V. PERFORMANCE EVALUATION

In this section, we compare the approximate solution in Section IV with the optimal but complicated cooperative strategy in Section II, as well as the literature.

A. Impact of User Number in Cooperation

One may wonder whether and how much an increasing number of cooperative users helps with an average user's expected privacy. Fig. 4 presents a numerical example for the expected privacy gain $\Pi^{appr} = \frac{1}{N} \sum_{u=1, \dots, N} \pi_u^{appr}$ of the approximate solution. To extend the second user's approximation strategy in Section IV-B to any user $i \in \{3, \dots, N\}$, here we fix with a reporting distance $r_{in} = \frac{1}{2}$ instead of the function r_{in}^* in (11) in the simulation. Notice that this is not an optimally designed strategy, yet it provides a lower bound performance for the optimal approximation solution. As N increases, Π^{appr} approaches the maximum possible privacy gain $\frac{1}{4}$, yet with a diminishing return. Asymptotically, we can prove that former cooperative users tend to cover all the location set for latter users to relax the service constraints, i.e., $\mathcal{X}_N^{IN}(x_{N-1}') \xrightarrow{a.s.} [0, 1]$. In this case, user i at a large enough order (with probability $\frac{1}{N}$) can always take advantage of former users' reports to relax the service constraint. Hence

an average user can achieve the maximum possible privacy gain, as long as there are enough number of cooperative users.

Proposition 3. *The expected privacy gain for an average user over any cooperation order increases with the number N of cooperative users. When $N \rightarrow \infty$, a user's expected privacy gain Π^{appr} averaging all cooperative orders under the approximate solution reaches the maximum $\frac{1}{4}$.*

B. Analytical Comparison with the Optimum

To compare our approximate solution in Section IV with the optimal solution from Problem (7) or (8), we first give an upper bound for the performance loss, then we numerically solve the optimal approach to further compare.

Let Π^{opt} denote an average user's expected privacy gain obtained by (1) and Problem (7). To explore how much privacy loss by using our approximate solution, we consider the maximum privacy gain $\Pi^{opt} \leq \frac{1}{4}$ from Proposition 1.

Proposition 4. *For an average user at any cooperation order randomly, the approximate reporting strategy reaches at least $\frac{3}{5}$ of the maximum privacy gain in Problem (7).*

Notice that this upper bound is derived using the maximum possible privacy value $\frac{1}{4}$, rather than the optimal one. Hence the exact performance loss of our approximate solution should be much smaller than 40%. This provides a lower bound for the efficiency and guaranteed performance of the approximate cooperative privacy-preserving scheme in Section IV. The gap between Π^{opt} and Π^{appr} in Fig. 4 also shows the performance loss of our approximation, which is smaller as N increases.

C. Comparison with Related Work

For a fair comparison, we first consider the privacy-preservation approaches with single query. Fig. 4 shows that when only allowing the user to query once to the LBS server, our approximate solution under the cooperative scheme is much better than the single-query cooperative approach with real location in [5]. Recall that the cooperative approach with k -anonymity scheme in [7]–[9] can achieve a better privacy preservation, yet with much overhead to users' mobile devices. We consider the unfair comparison with 2-anonymity in Fig. 4 and show that our cooperative approach with only one simple query still works well, with a low complexity and little overhead at the mobile devices.

VI. CONCLUSION

To address potential data leakage from LBS servers as well as peers, we study on multi-user privacy preservation problem under a cooperation scheme. We propose to leverage the user's service flexibility with one single query for a better privacy gain. Differently from the existing literature, we recommend users to query with misreported locations to the LBS server even if they find useful shared information, to jointly increase confusion to the adversary. To relax the complexity, we simplify the reporting scheme to a two-sided one, which shows significant privacy improvement compared with the related works.

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